

Cellular Biology and Biological Processes

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Executive Summary

This material covers fundamental concepts in cellular biology, including the structure and function of cells, photosynthesis, cellular respiration, and genetic processes. It also explores cell transport mechanisms, enzyme activity, homeostasis, and human physiological systems.

KEY TAKEAWAYS

- The cell is the basic structural and functional unit of all living organisms, and every organism is composed of one or more cells.
- Photosynthesis converts light energy into chemical energy in the chloroplast, producing glucose and oxygen from carbon dioxide and water.
- Cellular respiration involves glycolysis, the Krebs cycle, and the electron transport chain to produce ATP from glucose and oxygen.
- DNA carries genetic instructions, and protein synthesis involves transcription and translation to build proteins.
- Cell division includes interphase, mitosis, and cytokinesis, producing genetically identical daughter cells.
- Genetics explains inheritance through genes, alleles, genotypes, and phenotypes, with Punnett squares predicting trait probabilities.
- Enzymes act as biological catalysts, with active sites binding substrates and optimal temperatures enhancing their function.
- Homeostasis maintains stable internal conditions through mechanisms like negative feedback and hormone regulation.

Topics

TOPIC 1

Cellular Biology

The cell is the basic structural and functional unit of all living organisms, and the cell theory underpins all biological organization.

Cell Theory

- The cell theory states that all living things are composed of cells.
- New cells are produced from existing cells.

Cell Structure

- The nucleus is the membrane-bound organelle that contains the genetic material.
- The mitochondria is the organelle responsible for producing ATP through cellular respiration.

Cell Composition

- Every organism is composed of one or more cells.

TOPIC 2

Photosynthesis

Photosynthesis is the process by which green plants transform light energy into chemical energy, occurring in the chloroplast.

Process Overview

- Photosynthesis is defined as the process by which green plants transform light energy into chemical energy.

Location and Pigments

- Photosynthesis occurs in the chloroplast, which contains the green pigment chlorophyll.
- Chlorophyll is the pigment that absorbs light energy.

Chemical Equation

- The overall equation for photosynthesis shows that carbon dioxide and water, in the presence of light, produce glucose and release oxygen.
- The summary equation of photosynthesis is $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Reaction Stages

- The light-dependent reactions occur in the thylakoid membranes and produce ATP and NADPH.
- The Calvin cycle is the light-independent stage that fixes carbon dioxide into glucose.

TOPIC 3

Cellular Respiration

Cellular respiration consists of three main stages: glycolysis, the Krebs cycle, and the electron transport chain.

Stages of Respiration

- Glycolysis occurs in the cytoplasm and is the first stage, during which glucose is broken down into two molecules of pyruvate.
- The Krebs cycle occurs in the mitochondrial matrix.
- The electron transport chain is the final stage and occurs on the inner mitochondrial membrane.

Anaerobic Process

- Fermentation is the anaerobic process that regenerates NAD^+ when oxygen is unavailable.

Chemical Equation

- The summary equation of cellular respiration is $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$.

TOPIC 4

Cell Transport

The cell membrane is a selectively permeable barrier that controls the movement of substances in and out of the cell.

Passive Transport

- Diffusion is the passive movement of molecules from an area of high concentration to low concentration.

Water Diffusion

- Osmosis is defined as the diffusion of water across a selectively permeable membrane.

Active Transport

- Active transport is the movement of substances against their concentration gradient, requiring energy in the form of ATP.

Bulk Transport

- Endocytosis is the process by which the cell engulfs large particles by wrapping the membrane around them.

TOPIC 5

DNA and Protein Synthesis

DNA carries genetic instructions, and protein synthesis involves transcription and translation to build proteins.

DNA Structure and Function

- DNA is defined as the molecule that carries the genetic instructions for the development and functioning of living organisms.

DNA Replication

- DNA replication is the process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules.

Transcription

- Transcription is the process by which the DNA sequence of a gene is copied into messenger RNA.

Translation

- Translation is defined as the process by which the ribosome builds a protein from the mRNA sequence.
- Each three-nucleotide codon on mRNA specifies one amino acid.

Genetic Code

- The genetic code is the set of rules by which information encoded in genetic material is translated into proteins.

TOPIC 6

Cell Division

Cell division includes interphase, mitosis, and cytokinesis, producing genetically identical daughter cells.

Cell Cycle

- The cell cycle consists of interphase, mitosis, and cytokinesis.

Interphase

- Interphase is the stage during which the cell grows, copies its DNA, and prepares for division.

Mitosis Stages

- Mitosis is the process of nuclear division that produces two genetically identical daughter cells.
- Prophase is the first stage of mitosis, during which chromosomes condense.
- Anaphase is the stage during which sister chromatids are pulled apart to opposite poles of the cell.

TOPIC 7

Genetics

Genetics explains inheritance through genes, alleles, genotypes, and phenotypes, with Punnett squares predicting trait probabilities.

Gene and Allele

- A gene is defined as the basic unit of heredity, a segment of DNA that codes for a specific trait.
- An allele is an alternative form of a gene at the same locus.

Genotype and Phenotype

- The genotype is the genetic makeup of an organism, while the phenotype is its observable characteristics.

Allele Expression

- A dominant allele is expressed when only one copy is present.
- A recessive allele is only expressed when two copies are present.

Punnett Squares

- Punnett squares are used to predict the probability of offspring inheriting particular traits.

TOPIC 8

Enzymes

Enzymes act as biological catalysts, with active sites binding substrates and optimal temperatures enhancing their function.

Enzyme Function

- An enzyme is defined as a protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.

Active Site and Substrate Binding

- The active site is the region of an enzyme where the substrate binds.
- The induced fit model describes how the enzyme changes shape to accommodate the substrate.

Enzyme Activity Factors

- Optimal temperature is the temperature at which an enzyme works fastest.

Enzyme Denaturation

- Denaturation is defined as the loss of an enzyme's three-dimensional shape, which destroys its function.

TOPIC 9

Homeostasis

Homeostasis is the maintenance of a stable internal environment despite external changes, regulated by feedback mechanisms.

Feedback Mechanisms

- Negative feedback is a mechanism that reverses a change, bringing the system back to its set point.
- Positive feedback amplifies a change and drives the system further from its set point.

Human Body Examples

- The normal human body temperature is maintained at approximately 37 degrees Celsius.
- Blood glucose concentration is regulated by the hormones insulin and glucagon.
- Insulin is secreted when blood glucose is high, promoting glucose uptake.

TOPIC 10

Human Systems

The circulatory, respiratory, nervous, and endocrine systems work together to maintain physiological functions.

Circulatory System

- The circulatory system transports oxygen, nutrients, and waste products throughout the body.
- The heart is the muscular organ that pumps blood through the circulatory system.

Respiratory System

- The respiratory system is responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide.

Nervous System

- The nervous system coordinates rapid responses through electrical signals called nerve impulses.

Endocrine System

- The endocrine system regulates long-term processes through chemical messengers called hormones.

Terms & Definitions

Term	Definition
cell	The basic structural and functional unit of all living organisms.
cell theory	A fundamental principle of biology that states that all living things are composed of cells and that new cells are produced from existing cells.
nucleus	The membrane-bound organelle that contains the genetic material.
mitochondria	The organelle responsible for producing ATP through cellular respiration.
photosynthesis	The process by which green plants transform light energy into chemical energy.
chloroplast	The organelle where photosynthesis occurs, containing the green pigment chlorophyll.
chlorophyll	The pigment that absorbs light energy during photosynthesis.
glucose	The summary equation of photosynthesis shows that carbon dioxide and water, in the presence of light, produce glucose and release oxygen.
C₆H₁₂O₆	The summary equation of photosynthesis is $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.
light-dependent reactions	The reactions that occur in the thylakoid membranes and produce ATP and NADPH.
Calvin cycle	The light-independent stage of photosynthesis that fixes carbon dioxide into glucose.
cellular respiration	The process by which cells produce energy in the form of ATP through the breakdown of glucose.
glycolysis	The first stage of cellular respiration, occurring in the cytoplasm, where glucose is broken down into two molecules of pyruvate.
Krebs cycle	The stage of cellular respiration that occurs in the mitochondrial matrix.
electron transport chain	The final stage of cellular respiration, occurring on the inner mitochondrial membrane.
fermentation	The anaerobic process that regenerates NAD ⁺ when oxygen is unavailable.
cell membrane	The selectively permeable barrier that controls what enters and leaves the cell.
diffusion	The passive movement of molecules from an area of high concentration to low concentration.
osmosis	The diffusion of water across a selectively permeable membrane.
active transport	The movement of substances against their concentration gradient, requiring energy in the form of ATP.

Term	Definition
endocytosis	The process by which the cell engulfs large particles by wrapping the membrane around them.
DNA	The molecule that carries the genetic instructions for the development and functioning of living organisms.
DNA replication	The process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules.
transcription	The process by which the DNA sequence of a gene is copied into messenger RNA.
translation	The process by which the ribosome builds a protein from the mRNA sequence.
genetic code	The set of rules by which information encoded in genetic material is translated into proteins.
mitosis	The process of nuclear division that produces two genetically identical daughter cells.
cell cycle	The cycle that consists of interphase, mitosis, and cytokinesis.
interphase	The stage during which the cell grows, copies its DNA, and prepares for division.
prophase	The first stage of mitosis, during which chromosomes condense.
anaphase	The stage of mitosis during which sister chromatids are pulled apart to opposite poles of the cell.
gene	The basic unit of heredity, a segment of DNA that codes for a specific trait.
allele	An alternative form of a gene at the same locus.
genotype	The genetic makeup of an organism.
phenotype	The observable characteristics of an organism.
dominant allele	An allele expressed when only one copy is present.
recessive allele	An allele only expressed when two copies are present.
Punnett square	A tool used to predict the probability of offspring inheriting particular traits.
enzyme	A protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.
active site	The region of an enzyme where the substrate binds.
induced fit model	The model describing how the enzyme changes shape to accommodate the substrate.

Term	Definition
optimal temperature	The temperature at which an enzyme works fastest.
denaturation	The loss of an enzyme's three-dimensional shape, which destroys its function.
homeostasis	The maintenance of a stable internal environment despite external changes.
negative feedback	A mechanism that reverses a change, bringing the system back to its set point.
positive feedback	A mechanism that amplifies a change and drives the system further from its set point.
insulin	A hormone secreted when blood glucose is high, promoting glucose uptake.
glucagon	A hormone that regulates blood glucose concentration.
circulatory system	The system that transports oxygen, nutrients, and waste products throughout the body.
heart	The muscular organ that pumps blood through the circulatory system.
respiratory system	The system responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide.
nervous system	The system that coordinates rapid responses through electrical signals called nerve impulses.
endocrine system	The system that regulates long-term processes through chemical messengers called hormones.
ATP	The mitochondria is the organelle responsible for producing ATP through cellular respiration.
NADPH	The light-dependent reactions occur in the thylakoid membranes and produce ATP and NADPH.
NAD⁺	Fermentation is the anaerobic process that regenerates NAD ⁺ when oxygen is unavailable.
equation of cellular respiration	The summary equation of cellular respiration is $C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$.
equation of photosynthesis	The summary equation of photosynthesis is $6CO_2 + 6H_2O = C_6H_{12}O_6 + 6O_2$.

Key Facts & Formulas

Formula / Fact	Context
respiration is $C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$.	respiration is $C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$.
$C_6H_{12}O_6$	produce glucose and release oxygen. The summary equation of photosynthesis is $6CO_2 + 6H_2O = C_6H_{12}O_6 + 6O_2$. The light-dependent reactions occur in the thylakoid membranes and produce ATP and